

Lab 3: Photovoltaic Devices

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This lab investigates the behavior of various photovoltaic devices. Materials used include a small signal diode, a silicon photovoltaic device, one BPW34 photodiode, an LED light, and a cardboard box. A Keysight U2722A SMU was utilized to quickly collect data.

The BPW34 photodiode being used is a silicon semiconductor and contains three main layers. A p-region, an intrinsic layer in the middle, and an n-region. The small window takes in light, and once enough energy is absorbed, electron-hole pairs are created. In reverse bias, the depletion region, where those pairs are created, gets larger. This leads to less capacitance and a faster speed, meaning it is easier to produce a photocurrent. Where there is no bias, the device will still produce a current with a given light, but it will take longer. These small devices are typically used as small sensors in various electrical instruments. Most smoke detectors used them to determine a rapid change in light.

A silicon solar cell is similar to the BPW34, just larger. The main function is focused on producing an output rather than determining a change in its environment. The solar cell has a similar structure and power generation process to the photodiode. There are multiple layers, a front and back contact point, an intrinsic layer, and the p/n regions, respectively. Captured light creates electron-hole pairs in the depletion region and the electrons move to the n-side while the holes migrate to the p-side. The photovoltage is created through this carrier difference. The most

common use for a solar cell is in large scale solar farms. Massive amounts of panels capture the sunlight and turn it into power for what is required.

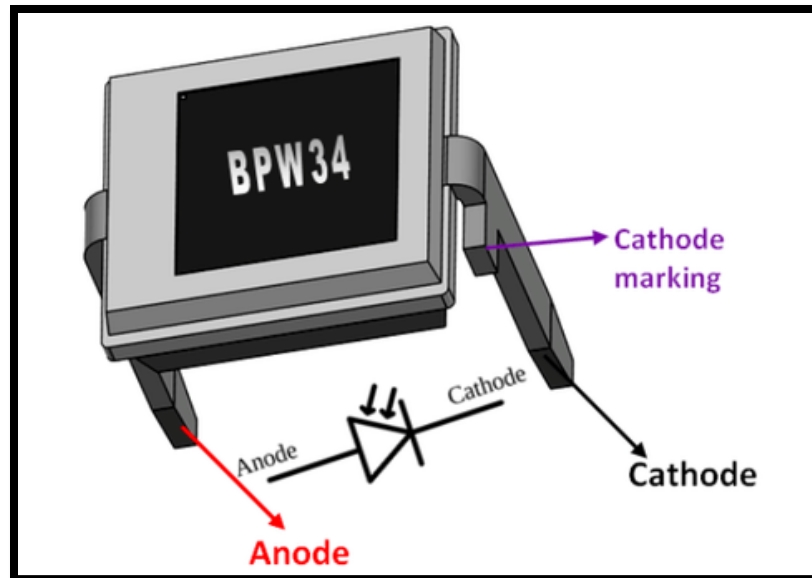


Figure 1: BPW34 Photodiode

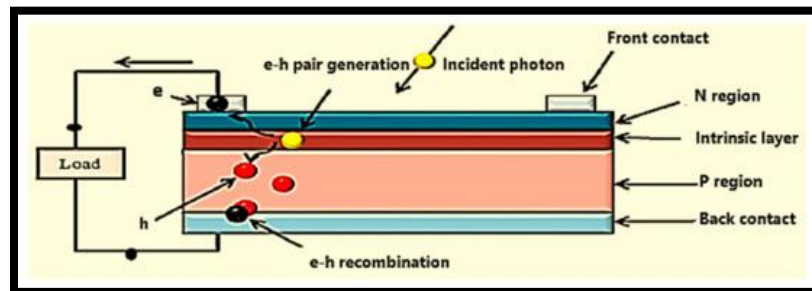


Figure 2: Solar Cell Structure

3A Fundamentals:

As similarly explained above, a photovoltaic device operates by utilizing captured light and creating energy. The trapped photons create electron-hole pairs that are eventually separated. When connected to a separate circuit, the charge flows through the load providing power. The short-circuit current, I_{SC} , is the maximum possible current that can be created when the device's leads are shorted. The open-circuit voltage, V_{OC} , is the maximum possible voltage created when there is no current. The fill factor, FF, is how effectively, not efficiently, the photovoltaic device

converts light into power. The fill factor usually gauges the quality of the diode. Lastly, the maximum power point, V_{MPP} , is the point on the device's I-V curve where the maximum current and voltage reside. The peak operating point of the device. Its power conversion efficiency is simply calculated by taking the device's maximum output and dividing it by the solar power it absorbed.

For the data analysis of a sum-squared error of a curve fit, data from Lab 2 was taken. Specifically, when measurements from a green LED were fit to the non-ideal diode equation.

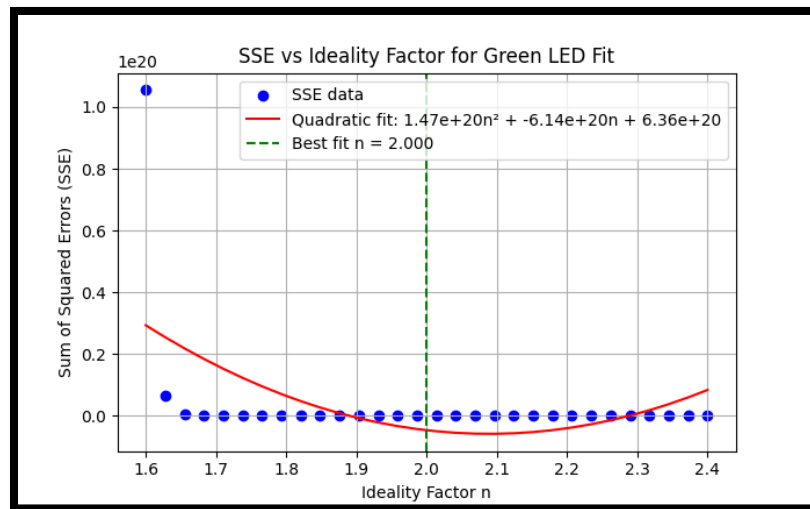


Figure 3: SSE vs Ideality Factor n

Experiment 1: Precision Diode I-V Measurement with an SMU

The goal of this experiment is to learn how to use the Keysight U2722A. This was accomplished by learning the software and running a simple I-V scan. The SMU is connected to the lab computer and a DC adapter. The Agilent Measurement Manager software is configured to collect data. The terminal blocks are connected to the output channels and the S^+ and O^+ as well as the S^- and O^- channels are connected with a short wire too. Simply because not all four channels are needed for this lab. The O^+ and O^- terminals are connected to the breadboard to a small signal diode. Using the SMU, an I-V scan is run. A second scan was run once we realized

how quickly the SMU can record data points. The second scan contains 80 data points. The data is plotted below with I_0 and n being a free parameter. To ensure the non-ideal fit appears close to the plotted data, only positive points were chosen for the fit. Furthermore, to prevent a negative R^2 value, multiple values were calculated depending on the points used. It should be noted, however, that the R^2 value of 0.7347 is the best to use because it explains how the non-ideal fit, with free parameters I_0 and n , fits the measured data most accurately.

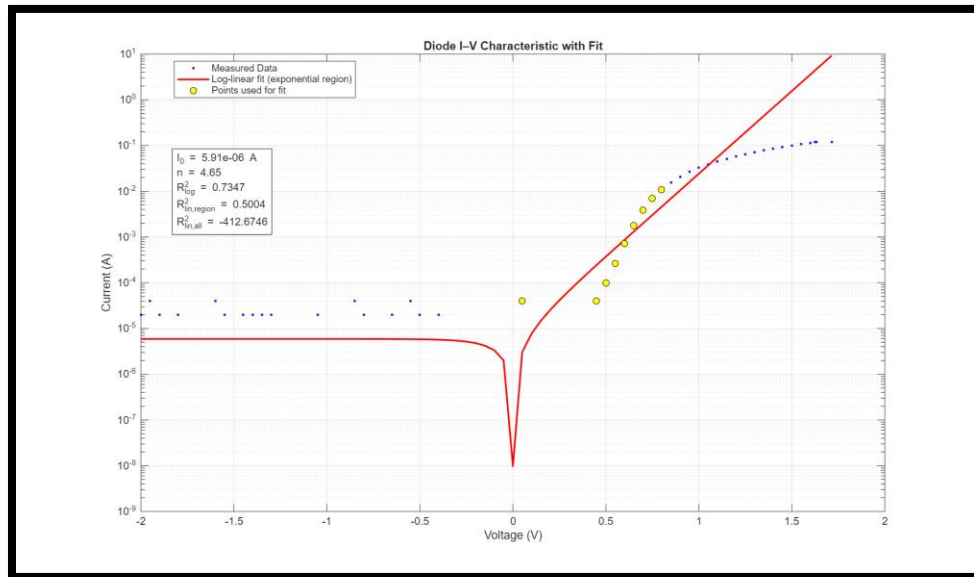


Figure 4: Diode I-V Curve with Non-Ideal Fit

The I_0 value was directly measured to be $-0.0000000078\text{A} = -0.0078\mu\text{A}$.

Concept Questions:

When utilizing points that help fit the non-ideal diode equation to the actual gathered data, the measurements line up excellently. Although not an exact fit, it is clear to see the similarities in behavior. The calculated free parameter I_0 value was 0.00000591A while the directly measured value comes out to be -0.0000000078A . Comparing the two, the calculated I_0 does not match perfectly, but it still is reasonable enough to produce a good fit.

The advantages of using the SMU are outstanding. 80 data points were able to be taken with minimal effort. If having to manually change the value of a power supply like in previous labs, it would have taken hours. The instrument is invaluable when it comes to taking large scale data points.

Experiment 2: Measuring the I-V Properties of a Solar Cell in the Dark

The goal of experiment 2 is to measure the I-V curve of a solar cell in the dark. The setup is similar to experiment 1. The SMU is connected to the solar cell and is covered with a box to block any light from reaching it. A sweep is run from 0 V to just under the turn on voltage. In this case, that value is 1.5V. This data is taken and plotted to measure the I-V curve.

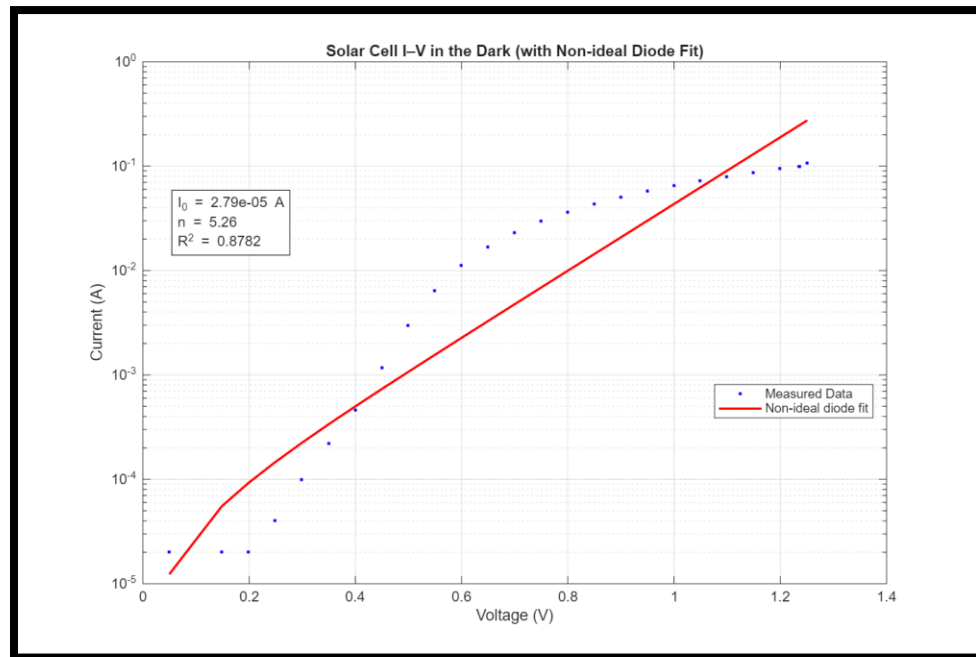


Figure 5: Solar Cell I-V Curve with Non-Ideal Fit

The I_0 value is $\sim 2.79 \times 10^{-5}$, the n value is ~ 5.26 , and the R^2 value is ~ 0.8782 .

Concept Questions:

The behavior fits well the non-ideal fit. It behaves very similarly to previous tested diodes from Lab 1. The curve also resembles a few of the LEDs tested in Lab 2. It fits better with the non-ideal fit than it would with the ideal fit. The measurements show the solar cell starts to taper off instead of getting increasingly exponential.

Experiment 3: Measuring the I-V Properties of a Silicon Solar Cell Under Illumination

The goal of experiment 3 is to measure the I-V curve of a silicon solar cell absorbing light. The maximum power point, V_{MMP} , short circuit current, open circuit voltage, fill factor, and power conversion efficiency are calculated. The solar cell and the BPW34 photodiode are connected to a breadboard with the photodiode as close as possible to the center of the solar cell. The solar cell is connected the same as it was in experiment 2 with the photodiode connected to a different channel with a bias of -5 V. The cardboard box, previously being used to shield the circuit from light, is cut to fit a light source. In this case an LED flood light directly over the components. The flood light is powered with a constant current of 0.85A and a voltage of 12V. The I_{SC} and V_{OC} are measured as well as an I-V curve from a 0V bias to just above the measured V_{OC} .

The short circuit current, I_{SC} , is measured to be ~ -0.06562 A. The open circuit voltage, V_{OC} , is measured to be ~ 0.558 V. The I-V curve of the solar cell is plotted below:

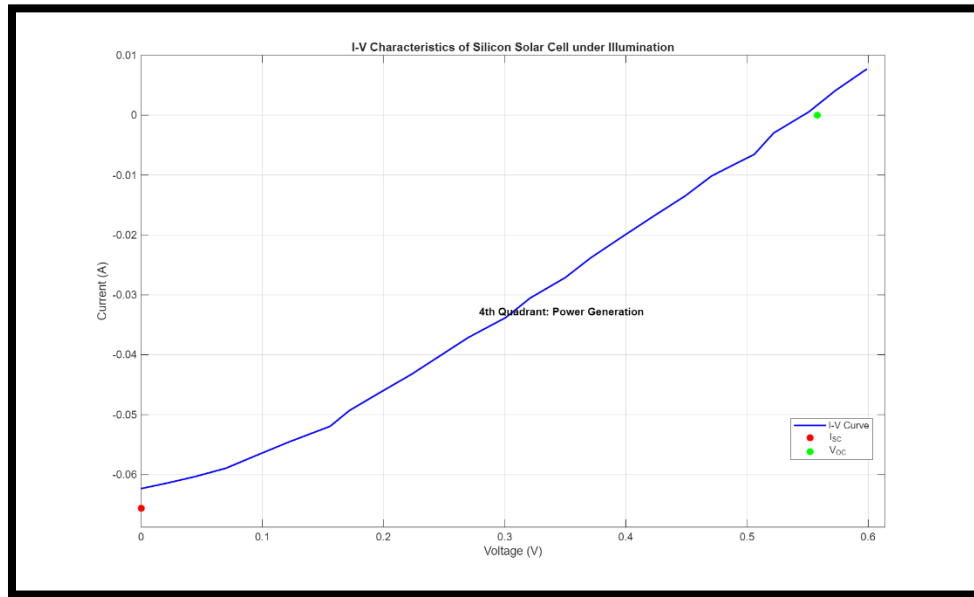


Figure 6: Solar Cell I-V Curve from 0V to V_{OC}

The current of the BPW34 photodiode, I_{PD} , is measured to be ~ -0.00102 A. The power generated is plotted and values for V_{MPP} , maximum power generated, and fill factor are calculated.

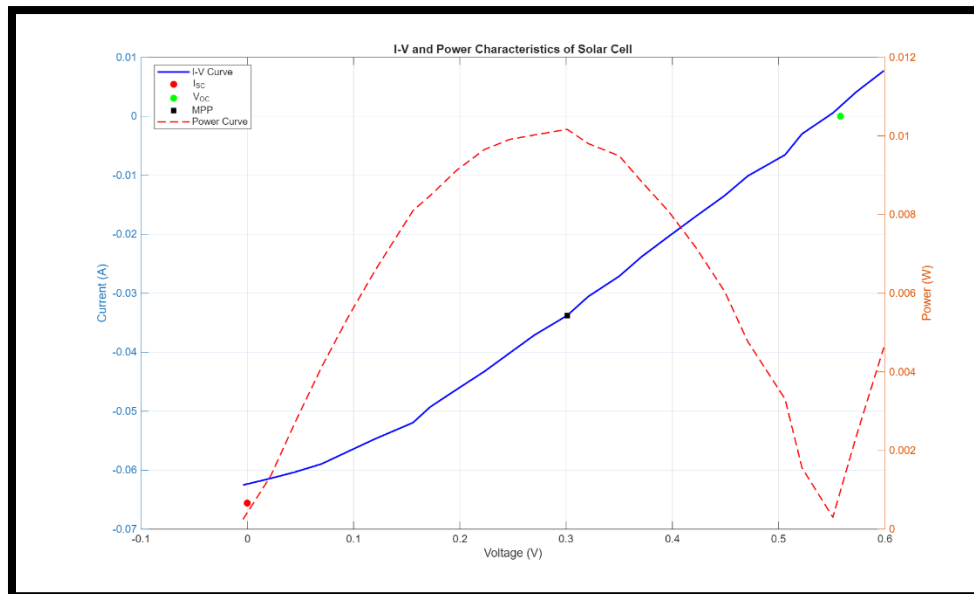


Figure 7: Power Characteristics

V_{MPP} is calculated to be ~ 0.301 V, I_{MPP} is calculated to be ~ -0.0338 A, the maximum power generated is calculated to be ~ 0.010168 W, and the fill factor is calculated to be $\sim 27.77\%$.

The irradiance of the solar cell is calculated as follows:

$$E = \frac{I_{PD}}{R} = \frac{1.02 \text{ mA}}{0.018} = 56.7 \text{ mW/cm}^2$$

Meaning the white LED produces about 57 mW/cm^2 of power to the surface of the solar cell.

Assuming the area of the solar cell is $1.5\text{cm} \times 5\text{cm}$ (*estimation as I didn't directly measure it*), the light power incident and efficiency can be calculated:

$$P_{in} = E \times A = 56.7 \text{ mW/cm}^2 \times 7.5\text{cm}^2 \approx 0.425\text{W} = \text{Light Incident}$$

$$\eta = \frac{P_{out}}{P_{in}} = \frac{0.010168\text{W}}{0.425\text{W}} = 0.02391 \approx 2.39\%$$

Concept Questions:

The light I-V curve of the solar cell is much more linear than the dark I-V curve. The light I-V curve resembles that of the dark but if it were moved down by the generated photocurrent. Current only flowed in the dark because of the diode's natural behavior following the non-ideal diode equation. But in the light, there is a constant photocurrent in the reverse direction. This is what creates the light I-V curve in the 4th quadrant. The maximum power point lies right in the middle of the I-V curve at around 0.301 V and the fill factor is around 27.77% . I would think V_{MPP} should be a little higher, but it is where the *power* is maximized, not just voltage. I would also assume that the fill factor should be higher. I expected something around at least 50% . 27.77% seems quite low, but with a low-level experiment like this, the illumination

certainly was not perfect, and the cell could have been quite hot by the time all meaningful data was measured. This could have contributed to the lower fill factor.

Experiment 4: Measuring the Light-Intensity Dependence of the Photovoltaic Behavior

The goal of experiment 4 is to measure the response of the solar cell as a function of its multiple characteristics. The setup is the same as the previous and multiple values for the short circuit current and open circuit voltage is measured. Values were taken when $I_{LED} = 0.15A$, $0.20A$, $0.25A$, and $0.30A$. Plots for I_{SC} and V_{OC} are below:

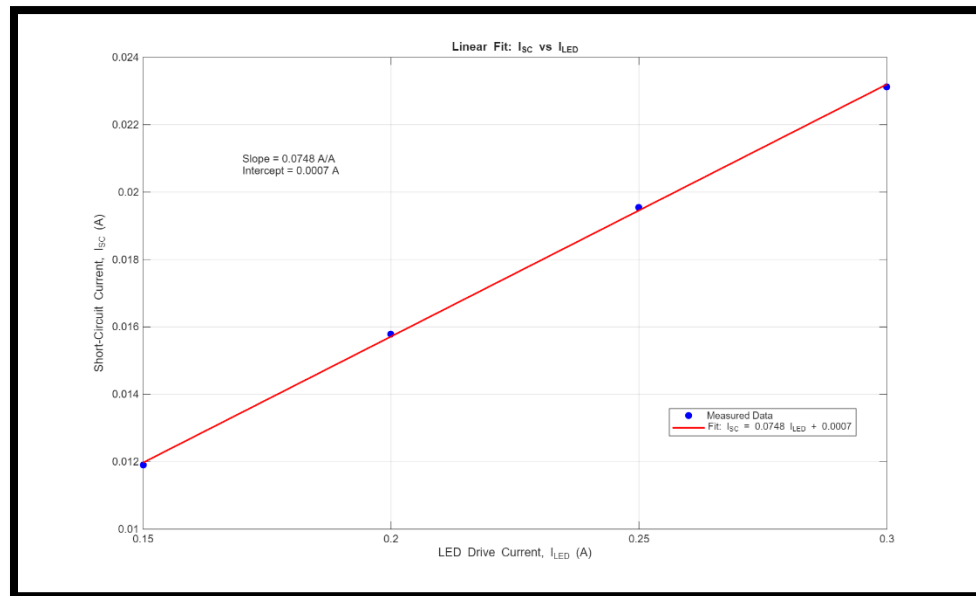


Figure 8: Linear fit of I_{SC} vs I_{LED}

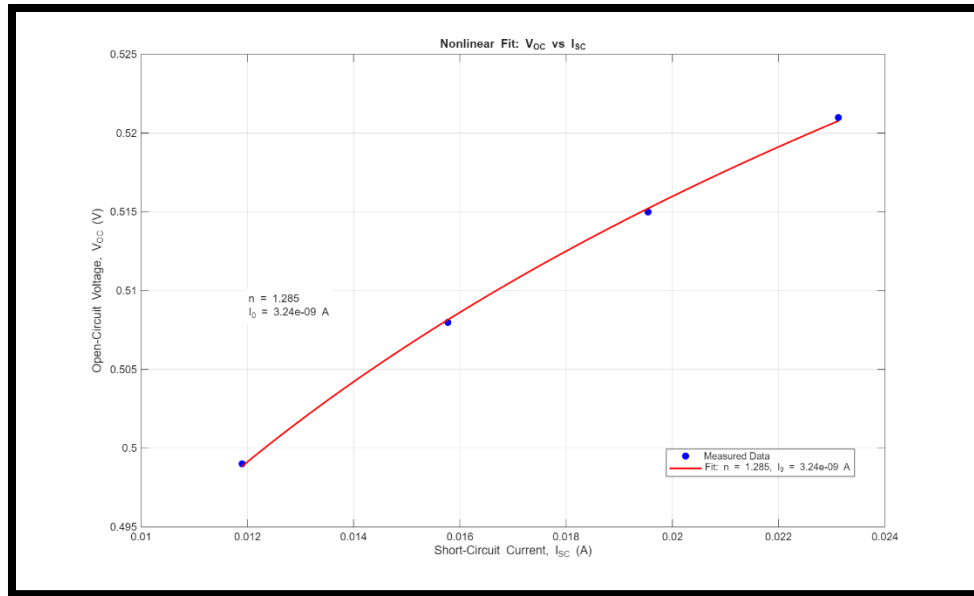


Figure 9: Nonlinear fit of V_{OC} vs I_{SC}

The n value for V_{OC} vs I_{SC} is 1.285 and I_0 is 3.24×10^{-9} A.

Concept Questions:

The fits of the solar cell look great compared to expected behavior. As the LED current increases, so do I_{SC} and V_{OC} values. The n and I_0 parameters are better when compared with the dark I-V curve. The n value is great but I_0 is still low compared to what I expected. However, both parameters are still well within reason.

3B Fundamentals:

$$I = I_0 \left(e^{\frac{qV_{OC}}{nk_bT}} - 1 \right) - I_L$$

$I = 0$ for open circuit and $V = V_{OC}$

$$I_L = I_0 \left(e^{\frac{qV_{OC}}{nk_bT}} - 1 \right)$$

$$e^{\frac{qV_{OC}}{nk_bT}} = \frac{I_L}{I_0}$$

Knowing: $J_{SC} = \frac{I_L}{A}$ and $J_0 = \text{Saturation density}$

$$V_{OC} = \frac{nk_bT}{q} \ln\left(\frac{J_{SC}}{J_0} + 1\right)$$

A solar cell's efficiency is largely determined by its own internal structure and material. The parameters I_{SC} , V_{OC} , V_{MPP} , and FF are what help directly calculate how efficient a specific cell might be. Not all absorbed photons get converted into electrical energy, further decreasing efficiency. The Shockley-Queisser limit is the absolute maximum theoretical efficiency and is determined to be about 32-33%. Each of the previous issues mentioned bring that value down more. The most common tradeoff is that a smaller band gap can absorb a larger number of photons (higher current) but will produce a lower voltage while a larger band can output more voltage but absorb less photons (less current). The Shockley-Queisser limit states that this tradeoff is best with a band gap of ~1.1-1.4 eV. Silicon has a band gap of 1.12 eV, which is why most solar cells are designed with Silicon.

Experiment 5: Measuring the Impact of Series Resistance on a Photovoltaic Device

The goal of experiment 5 is to understand how series resistance affects a photovoltaic device. The I-V curve for the photovoltaic device is measured when under the same LED flood light. Multiple resistors are placed in series with the device. The resistor values chosen can be seen in the table. The parameters I_{SC} , V_{OC} , FF, and efficiency η are calculated as function of R_s (Assuming an irradiance of 200 and solar cell area of 1.25cm x 5cm). The parameter values that were calculated are in the table below:

Resistor	1 Ω	10 Ω	100 Ω	470 Ω	1000 Ω	4700 Ω
I_{SC}	-0.036 A	-0.021 A	-0.005 A	-0.001 A	-0.0006 A	-0.0001 A
V_{OC}	0.57 V	0.56 V	0.56 V	0.54 V	0.58 V	0.55 V
FF	0.6916	0.4587	0.2498	0.2493	0.2465	0.3274
η	0.1141	0.0429	0.0060	0.0013	0.0007	0.0001

The I-V curves for the resistors are plotted:

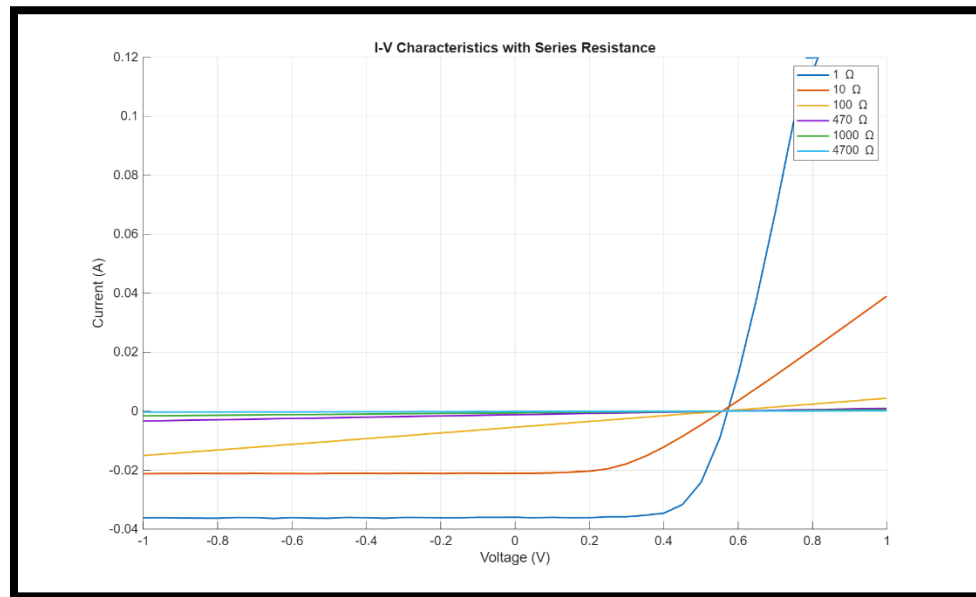


Figure 10: I-V Curves

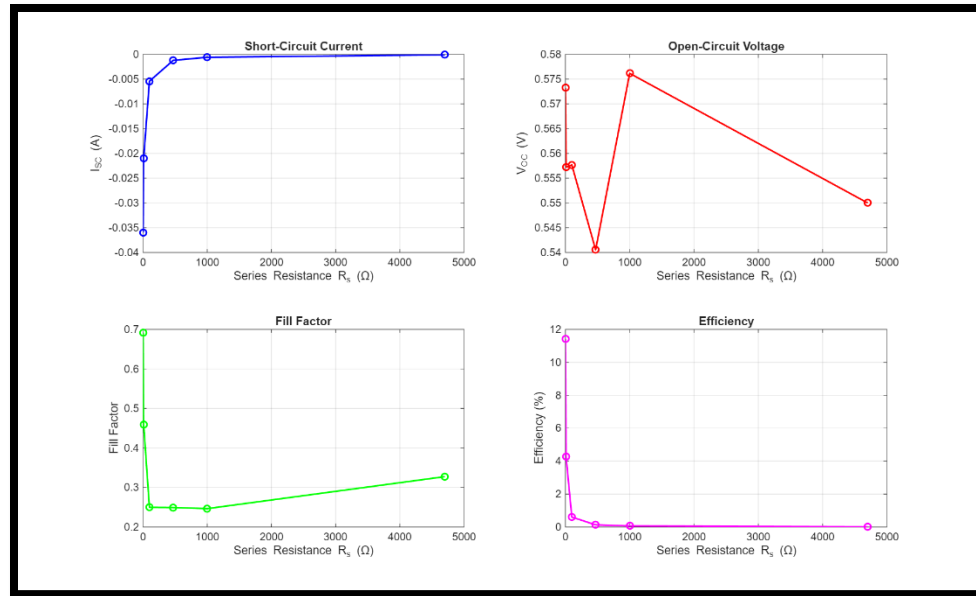


Figure 11: I_{sc} , V_{oc} , FF, and Efficiency η vs Series Resistance

Concept Questions:

The fill factor and efficiency η parameters appear to be most sensitive to R_s . For absolute best efficiency, R_s should be as close to 0 as possible. This is evident in the plot above. The larger the series resistance, the larger decrease in efficiency.

Experiment 6: Measuring the Impact of Shunt Resistance on a Photovoltaic Device

The goal of experiment 6 is to understand how shunt resistance affects a photovoltaic device. The I-V curve for the photovoltaic device is measured when under the same LED flood light. Multiple resistors are placed in parallel with the device. The resistor values chosen can be seen in the table. The parameters I_{SC} , V_{OC} , FF, and efficiency η are calculated as function of R_{SH} (Assuming an irradiance of 200 and solar cell area of 1.25cm x 5cm). The parameter values that were calculated are in the table below:

Resistor	1 Ω	47 Ω	220 Ω	470 Ω	1000 Ω	2200 Ω
I_{SC}	-0.04303 A	-0.04727 A	-0.04558 A	-0.03188 A	-0.04608 A	-0.04481 A
V_{OC}	0.04958 V	0.55467 V	0.57348 V	0.55873 V	0.57013 V	0.57117 V
FF	0.0159	0.5867	0.6999	0.7193	0.7299	0.7314
η	0.00027	0.12306	0.14635	0.10250	0.15339	0.14975

The I-V curves for the resistors are plotted:

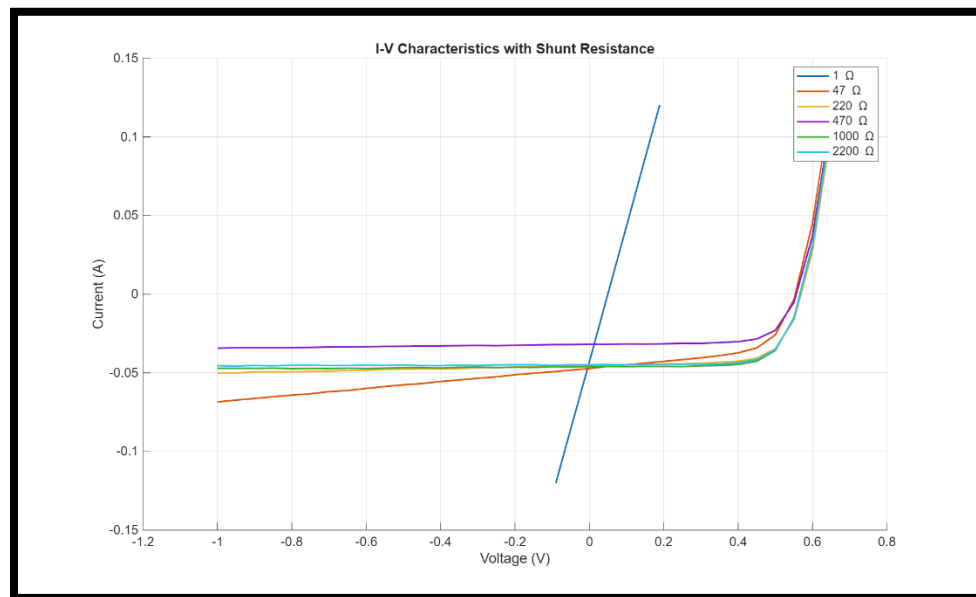


Figure 12: I-V Curves with Shunt Resistance

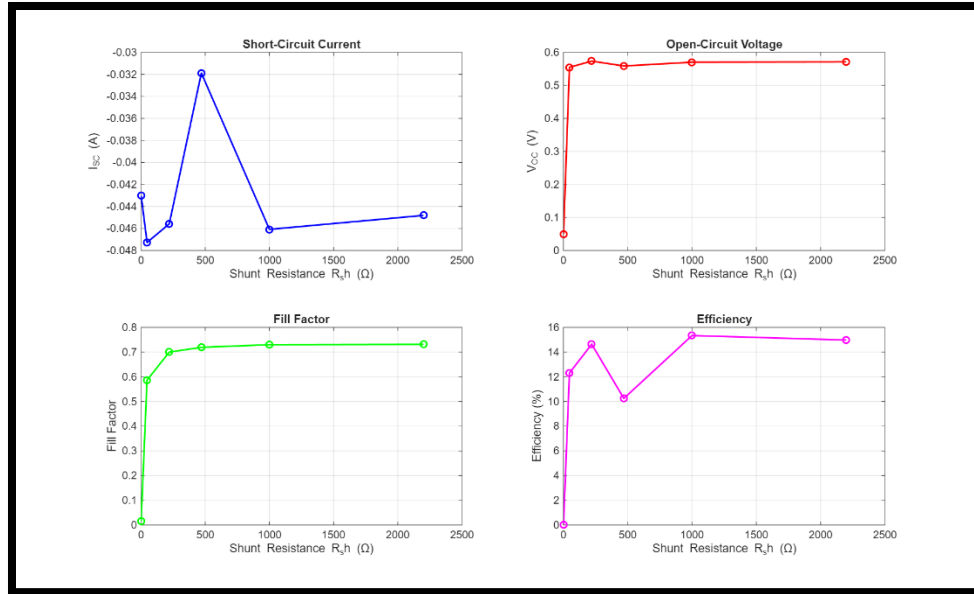


Figure 13: I_{sc} , V_{oc} , FF, and Efficiency η vs Shunt Resistance

The efficiency curve is taken and fit into the equivalent circuit model equation with a shunt resistor of 1000Ω .

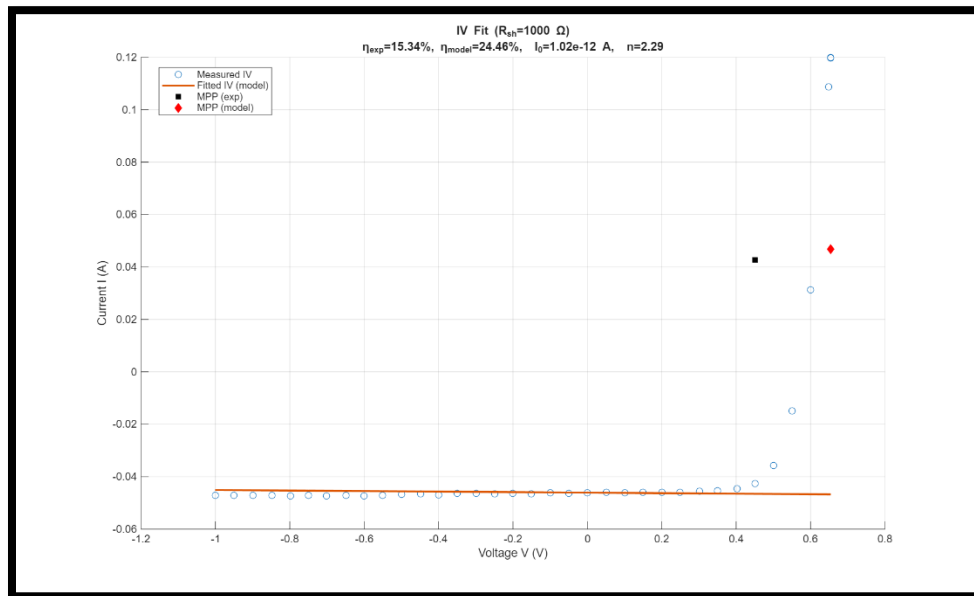


Figure 14: Efficiency η Fit to Equivalent Circuit Model with 1000Ω Shunt Resistance

I_0 is 1.02×10^{-12} A and n is 2.29.

Concept Questions:

The fill factor and efficiency η are the most sensitive to R_{sh} . The critical value for R_{sh} should be as large as possible. Opposite as to what was stated in experiment 5. A low R_{sh} will leak and bring down the fill factor and therefore efficiency η as well. This is evident in the parameter plots. The lower resistances have much smaller efficiencies. In the beginning, the fit is excellent when compared with the measured data. However, near the end, the measured data increases exponentially instead of remaining somewhat flat. Overall, the I_0 and n values appear to be within reason, so the fit does not necessarily seem out of the ordinary.

Lab 3A Simulation

The OghmaNano software is downloaded and installed to perform the following simulations. A silicon photovoltaic device is run with default parameters to observe the values of V_{OC} , J_{SC} , FF, and efficiency η . The simulation is run again with a doped n+ parameter. The new values of V_{OC} , J_{SC} , FF, and efficiency η are compared. *(Was told to ignore the plots of the band structure part).*

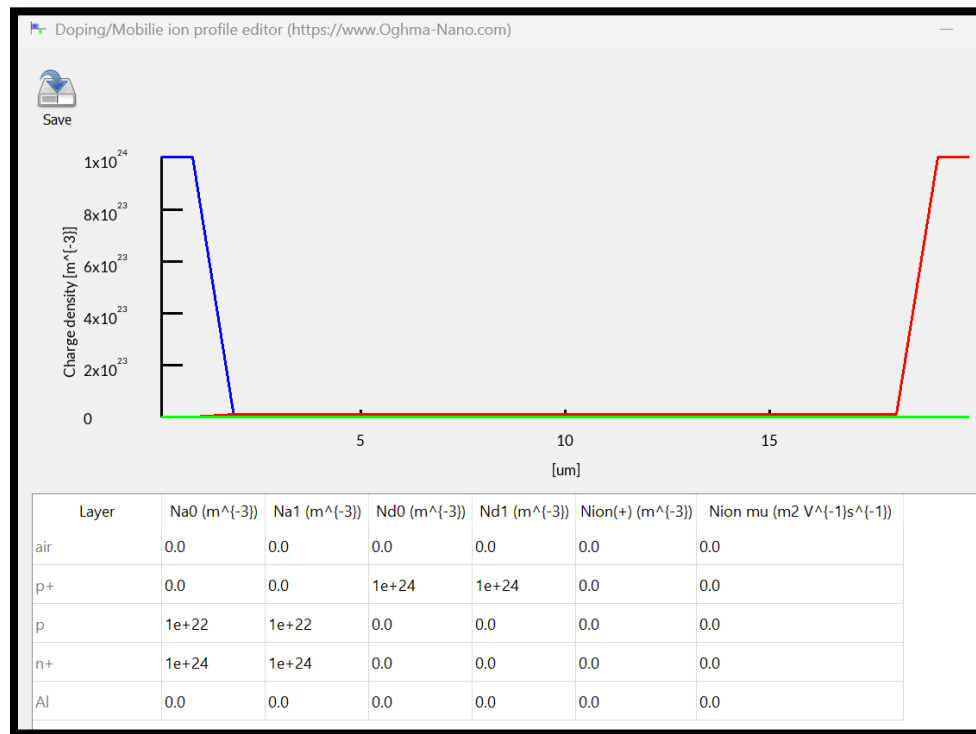


Figure 15: Default Parameters

Simulation information		
Edit	json_file	Icon
Current density at max power	-1.152457e+02	A m ⁻²
Voltage at max power	4.588475e-01	V
Fill factor	8.106016e-01	a.u.
Power conversion efficiency (PCE)	5.288020e+00	Percent
Max power	5.288020e+01	W m ⁻²
V _{oc}	5.463980e-01	V
J _{sc}	-1.193924e+02	A m ⁻²
Trapped electrons at Voc	0.000000e+00	m ⁻³
Trapped holes at Voc	0.000000e+00	m ⁻³
Free electrons at Voc	5.001245e+22	m ⁻³
Free holes at Voc	5.899963e+22	m ⁻³
V _{bi}	-1.600000e+00	V
Total carriers (n+p)/2 at Voc	5.450604e+22	au
Average mobility as J _{sc}	5.015000e+03	m ² V ⁻¹ s ⁻¹

Figure 16: Simulated V_{OC}, J_{SC}, FF, and Efficiency η for Default Parameters

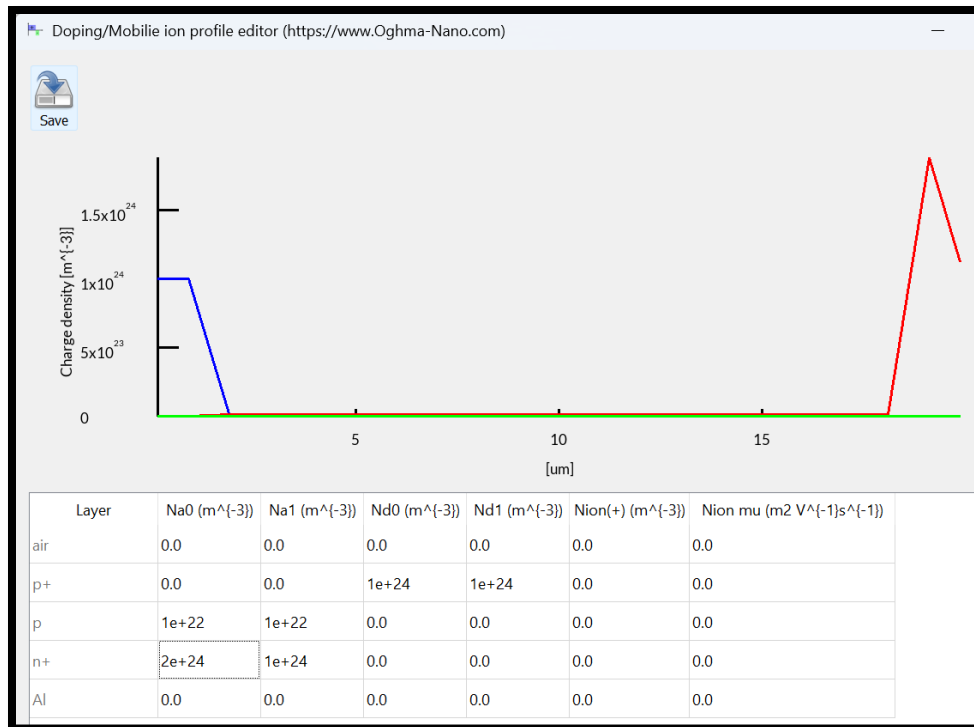


Figure 17: n⁺ Doped Parameters

Simulation information		
Edit	[icon] json_file	Icon
Current density at max power	-1.160372e+02	Am ⁻²
Voltage at max power	4.488396e-01	V
Fill factor	7.983868e-01	a.u.
Power conversion efficiency (PCE)	5.208208e+00	Percent
Max power	5.208208e+01	W m ⁻²
V _{oc}	5.464499e-01	V
J _{sc}	-1.193781e+02	A m ⁻²
Trapped electrons at Voc	0.000000e+00	m ⁻³
Trapped holes at Voc	0.000000e+00	m ⁻³
Free electrons at Voc	5.001245e+22	m ⁻³
Free holes at Voc	9.024965e+22	m ⁻³
V _{bi}	-1.600000e+00	V
Total carriers (n+p)/2 at Voc	7.013105e+22	au
Average mobility as J _{sc}	5.015000e+03	m ² V ⁻¹ s ⁻¹

Figure 18: Simulated V_{OC}, J_{SC}, FF, and Efficiency η for n+ Doped Parameters

	Default Parameters	Doped n+
V _{OC}	$5.463980 \times 10^{-1} \text{ V}$	$5.464499 \times 10^{-1} \text{ V}$
J _{SC}	$-1.193924 \times 10^2 \text{ Am}^{-2}$	$-1.193781 \times 10^2 \text{ Am}^{-2}$
FF	$8.106016 \times 10^{-1} \text{ a.u.}$	$7.983868 \times 10^{-1} \text{ a.u.}$
Efficiency η	5.288020%	5.208208%

Although slight, doping the n+ side decreased the FF and efficiency. V_{OC} and J_{SC} did change as well, but extremely slightly. If it were more heavily doped, the values would change significantly more. These changes are expected. Especially after performing the previous experiments, it is known that FF and η are the most sensitive to change. It would take an extreme amount of doping for V_{OC} and J_{SC} to drastically shift though.

Lab 3B Simulation

The OghmaNano software is downloaded and installed to perform the following simulations. A silicon photovoltaic device is run with default parameters to observe the values of V_{OC} , J_{SC} , FF, and efficiency η . The simulation is run again with a significantly doped hole parameter. The new values of V_{OC} , J_{SC} , FF, and efficiency η are compared. *(Was told to ignore the plots of the band structure part).*

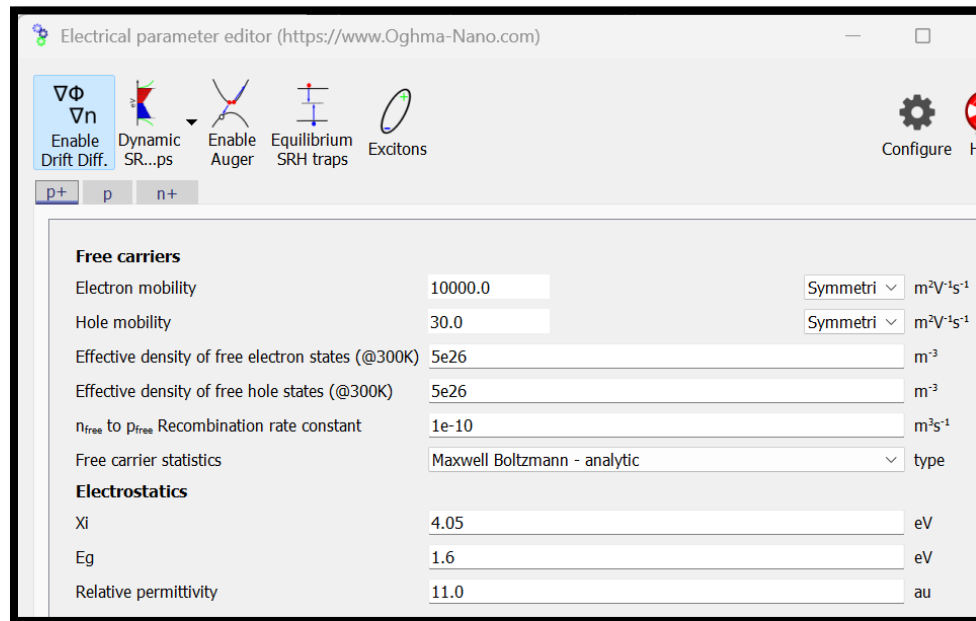


Figure 19: Default Parameters

Simulation information		
Edit	json_file	Icon
Current density at max power	-1.152457e+02	$A m^{-2}$
Voltage at max power	4.588475e-01	V
Fill factor	8.106016e-01	a.u.
Power conversion efficiency (PCE)	5.288020e+00	Percent
Max power	5.288020e+01	$W m^{-2}$
V_{oc}	5.463980e-01	V
J_{sc}	-1.193924e+02	$A m^{-2}$
Trapped electrons at Voc	0.000000e+00	m^{-3}
Trapped holes at Voc	0.000000e+00	m^{-3}
Free electrons at Voc	5.001245e+22	m^{-3}
Free holes at Voc	5.899963e+22	m^{-3}
V_{bi}	-1.600000e+00	V
Total carriers $(n+p)/2$ at Voc	5.450604e+22	au
Average mobility as J_{sc}	5.015000e+03	$m^2V^{-1}s^{-1}$

Figure 20: Simulated V_{OC} , J_{SC} , FF, and Efficiency η for Default Parameters

Electrical parameter editor (<https://www.Oghma-Nano.com>)

☒ $\nabla\Phi$ ∇n Enable Drift Diff.
 ☒ Dynamic SR...ps
 ☒ Enable Auger
 ☒ Equilibrium SRH traps
 ☒ Excitons
 ☐ Configure

p+ p n+

Free carriers

Electron mobility	10000.0	Symmetri	$\text{m}^2\text{V}^{-1}\text{s}^{-1}$
Hole mobility	150	Symmetri	$\text{m}^2\text{V}^{-1}\text{s}^{-1}$
Effective density of free electron states (@300K)	5e26		m^{-3}
Effective density of free hole states (@300K)	5e26		m^{-3}
n_{free} to p_{free} Recombination rate constant	1e-10		m^3s^{-1}
Free carrier statistics	Maxwell Boltzmann - analytic		type

Electrostatics

χ	4.05	eV
E_g	1.6	eV
Relative permittivity	11.0	au

Figure 21: Doped Hole Mobility Parameters

Configure (<https://www.Oghma-Nano.com>)

Simulation information

Edit	{:} json_file	Icon
Current density at max power	-1.190994e+02	A m^{-2}
Voltage at max power	4.788090e-01	V
Fill factor	8.208417e-01	a.u.
Power conversion efficiency (PCE)	5.702589e+00	Percent
Max power	5.702589e+01	W m^{-2}
V_{oc}	5.470795e-01	V
J_{sc}	-1.269879e+02	A m^{-2}
Trapped electrons at V_{oc}	0.000000e+00	m^{-3}
Trapped holes at V_{oc}	0.000000e+00	m^{-3}
Free electrons at V_{oc}	5.001245e+22	m^{-3}
Free holes at V_{oc}	5.899963e+22	m^{-3}
V_{bi}	-1.600000e+00	V
Total carriers $(n+p)/2$ at V_{oc}	5.450604e+22	au
Average mobility as J_{sc}	5.018000e+03	$\text{m}^2\text{V}^{-1}\text{s}^{-1}$

Figure 22: Simulated V_{oc} , J_{sc} , FF, and Efficiency η for Doped Hole Mobility

	Default Parameters	Doped Hole Mobility
V_{OC}	$5.463980 \times 10^{-1} \text{ V}$	$5.4702589 \times 10^{-1} \text{ V}$
J_{SC}	$-1.193924 \times 10^2 \text{ Am}^{-2}$	$-1.269879 \times 10^2 \text{ Am}^{-2}$
FF	$8.106016 \times 10^{-1} \text{ a.u.}$	$8.208417 \times 10^{-1} \text{ a.u.}$
Efficiency η	5.288020%	5.702589%

Although slight, doping the hole mobility increased the FF and efficiency. V_{OC} and J_{SC} did change as well, but extremely slightly. V_{OC} hardly increased and J_{SC} hardly decreased. If the electron mobility was also doped, the values would change significantly more. These changes are expected. Especially after performing the previous experiments, it is known that FF and η are the most sensitive to change. It would take an extreme amount of doping for V_{OC} and J_{SC} to drastically shift though. Hole mobility would have to be massively increased while also have electron mobility either significantly increase or decrease.

Conclusion:

The general behavior of photovoltaic devices and how they function with varying light conditions and resistances was analyzed. I-V curves and efficiencies were measured for multiple conditions to understand their properties. I learned that, in general, solar cells and diodes are useful in multiple scenarios. A resistor in series with a solar cell affects the output the exact opposite than if it were in parallel. The efficiency of solar cells can be easily manipulated. My largest takeaway is that SMU is an invaluable tool when taking data points for experiments like these. It largely sped up the process of compiling data to through into I-V curve calculations. I learned useful knowledge of how to use the SMU and how solar cells function depending on light conditions and connected resistances.

Generative AI Usage:

AI was used to help with my plots and scale them correctly. With so many data points I could no longer manually enter all the data in Matlab. ChatGPT was the AI engine used for this as it explained how to import a .csv file. The prompts utilized were simple. I would provide my code from previous labs and ask it to make an edit to allow an import of a .csv file. It was able to combine multiple of my plots into four subplots allowing for better organization of my report. It helped sort out small discrepancies I missed when naming or scaling my plots too.

Citations:

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